HARI KIRAN VENTRAPRAGADA

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Backend and AI/ML engineer with experience across full-stack development, backend systems, and applied AI/LLM projects. Comfortable working across the stack — from system design and implementation to debugging and deployment. Strong foundation in Python, API design, and machine learning fundamentals, with a track record of independently owning features from concept to production.

EDUCATION

Maharaj Vijayaram Gajapathi Raj College of Engineering (MVGR), Autonomous *2022 – 2026*
B.Tech in Information Technology | CGPA: 7.76 / 10

EXPERIENCE

Software Development Intern — OmniqAI Pvt Ltd, Bengaluru *Dec 2025 – Jul 2026*

- Designed and implemented a real-time user presence system (online/away/offline/in-call states) for a chat platform using WebSockets; merged to production.
- Performed root-cause analysis (RCA) for production issues across chat, audio/video, and project-management modules — including duplicate-message rendering, AI-response rendering, and in-call presence desync — documenting findings and proposing fixes that were then implemented by the team; authored 10+ RCA documents in total.
- Contributed the React frontend and backend wiring for a Secrets & Keys management feature (for storing API keys/credentials) within the application; implementation completed, pending merge.
- Designed (architecture and technical research, not yet implemented) a Global Search feature spanning semantic and streaming search over chat data.

AI R&D Intern — OmniqAI Pvt Ltd, Bengaluru *May 2025 – Jun 2025*

- Contributed to an MCP (Model Context Protocol) Agent Orchestration proof-of-concept integrated with Mailcow, as part of the company's DevOps & AI Orchestration initiative, using FastAPI, Docker, and Playwright, with SonarQube for code quality checks.

AI Research Intern — OmniqAI Pvt Ltd, Bengaluru *Dec 2024 – Jan 2025*

- Built a Knowledge Graph microservice (Neo4j, Python, Ollama) to construct knowledge graphs from Slack conversational data as a grounding layer for a Retrieval-Augmented Generation (RAG) system, applying prompt engineering techniques (chain-of-thought, one-shot, few-shot) for dynamic querying.
- Containerized and deployed the service using Docker Swarm and Gitea CI; monitored with Prometheus.

PROJECTS

AI Calorie Tracker caltrack-bay.vercel.app | *Expo/React Native, FastAPI, Supabase, Gemini API*

- Built and deployed a cross-platform nutrition tracking app that uses the Gemini API to identify food and estimate calorie/macro content from photos and text descriptions.
- Designed the FastAPI backend and Supabase data layer to handle auth, logging, and structured nutrition data derived from LLM output.
- Built and shipped the entire stack on free-tier infrastructure end-to-end as a deliberate cost constraint, from mobile client to backend to database.

CareerPath AI | *LangGraph, FastAPI, Redis, MongoDB, React, AWS*

- Built a multi-agent career-advisory platform using LangGraph to orchestrate specialized agents for resume parsing, job matching, skill-gap analysis, and mock interviews.
- Implemented semantic matching and retrieval over resume and job-role data, with scoring and personalized roadmap generation.
- Used Redis and a message-driven architecture to coordinate agent tasks and manage state across a multi-step pipeline.
- Deployed the application on AWS (EC2, S3, RDS, Lambda) for compute, storage, and database infrastructure.

Autonomous Multi-Step AI Agent github.com/Harihkvent/Autonomous-Multi-Step-AI-Agent | *Python, FastAPI, LangGraph, React/Vite*

- Built a full-stack autonomous agent platform that decomposes natural-language tasks into multi-step plans and executes them through a tool registry (web search, calendar, email notifications, document parsing/generation).
- Implemented two execution architectures: a sequential Planner → Executor → Validator → Retry pipeline, and a multi-agent LangGraph supervisor that routes to specialized sub-agents (research, document handling, calculation, weather).
- Used an LLM for JSON-based task planning with heuristic fallbacks, and built a React/Vite frontend with real-time execution status updates.

DSA Visualizer | *React, TypeScript, TailwindCSS*

- Built an interactive web application visualizing 43+ data structures and algorithms through step-by-step execution, with custom inputs and playback controls.

TECHNICAL SKILLS

AI/ML: RAG, LangChain, LangGraph, Embeddings, Semantic Search, Prompt Engineering, NLP, Generative AI
Backend: Python, FastAPI, Node.js/Express, REST APIs, Microservices, Authentication
Databases: Neo4j, Redis, MongoDB, FAISS, SQL
Cloud/DevOps: Docker, AWS (EC2), Prometheus, GitHub Actions, Linux
Frontend: React, TypeScript, TailwindCSS, React Native
Languages: Python, Java, JavaScript, TypeScript, SQL, C++ **Tools:** Git, GitHub, Postman